Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A compound microscope has a magnification of 30. Assuming that the final image is formed at the least distance of distinct vision $(25\ \text{cm})$, find the magnification produced by the objective. Given, the focal length of the eyepiece is $5.0\ \text{cm}$.

(A) 4

(B) 5

(C) 6

(D) 8

Q.2

Find the magnifying power of a compound miscroscope whose objective and eyepiece are of focal lengths $4.0\ \text{cm}$ and $6.0\ \text{cm}$ respectively and the object is placed $5.0\ \text{cm}$ beyond the objective. Assume that the final image is formed at the least distance of distinct vision $(25\ \text{cm})$.

(A) -20.7

(B) 21.7

(C) -22.7

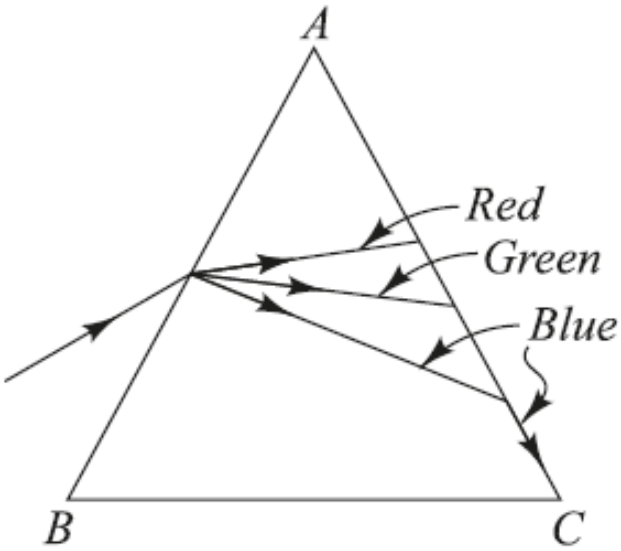
(D) -23.7

Q.3

A composite light consisting of red, green and blue colours is incidence on face AB of a glass prism. The

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blue light is just totally internally reflected at face AC as shown in Fig. 16.59. Then



(A) green light is internally reflected at AC but red light emerges from AC.

(B) red light is internally reflected at AC but green light emerges from AC

(C) green and red lights both emerge from AC

(D) green and red lights both do not emerge from AC

Q.4

A thin prism P_1 with angle 4° and made from glass of refractive index 1.54 is combined with another thin prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of prism P_2 is

(A) 5.33°

(B) 4°

(C) 3°

(D) 2.6°

Q.5

In the visible region, the dispersive powers and the mean angular deviations for crown and flint glass prisms are ω and ω' and d and d' respectively. When the two prism are combined, the condition of zero dispersion by the combination is

(A) $\sqrt{\omega d}+\sqrt{\omega' d'}=0$

(B) $\omega' d+\omega d'=0$

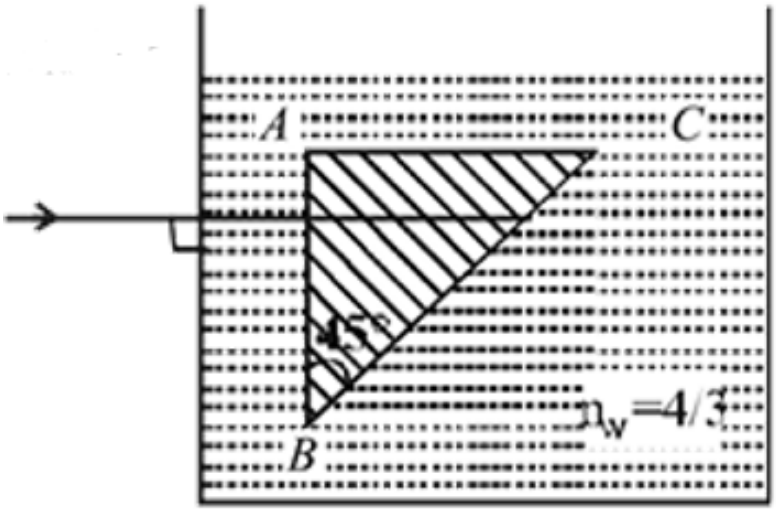
(C) $\omega d+\omega' d'=0$

(D) $(\omega d)^2+(\omega' d')^2=0$

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Q.6

A triangular prism of glass is inside water. A ray, incident normally, on one of the faces, is totally reflected from face BC. Then the minimum refractive index of glass is -A triangular prism of glass is inside water. A ray, incident normally, on one of the faces, is totally reflected from face BC. Then the minimum refractive index of glass is -



(A) $\frac{\sqrt{3}}{2}$

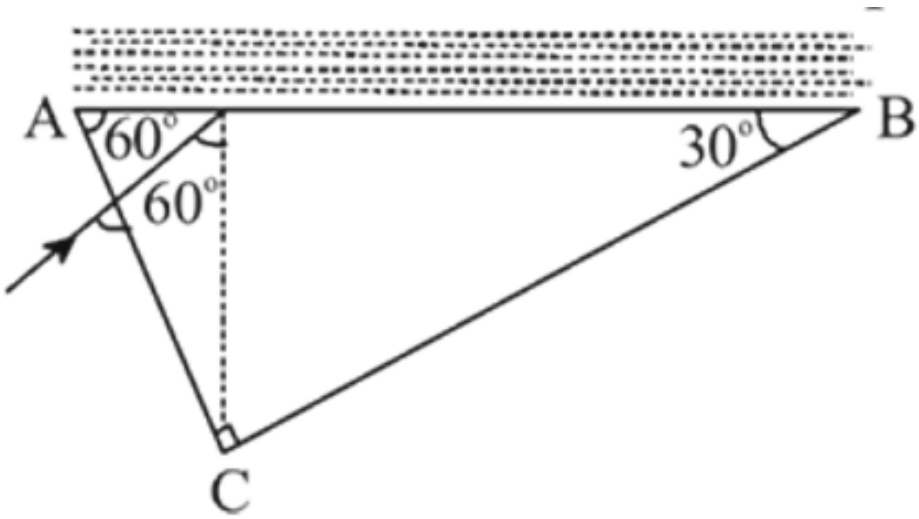
(B) $\frac{5}{3}$

(C) $\frac{2\sqrt{2}}{5}$

(D) $\frac{4\sqrt{2}}{3}$

Q.7

ACB is right - angle prism with other angles as 60° and 30°. Refractive index of the prism is 1.5. AB has thin layer of liquid on it as shown. Light falls normally on the face AC. For total internal reflections, maximum refractive index of the liquid is



(A) 1.4

(B) 1.3

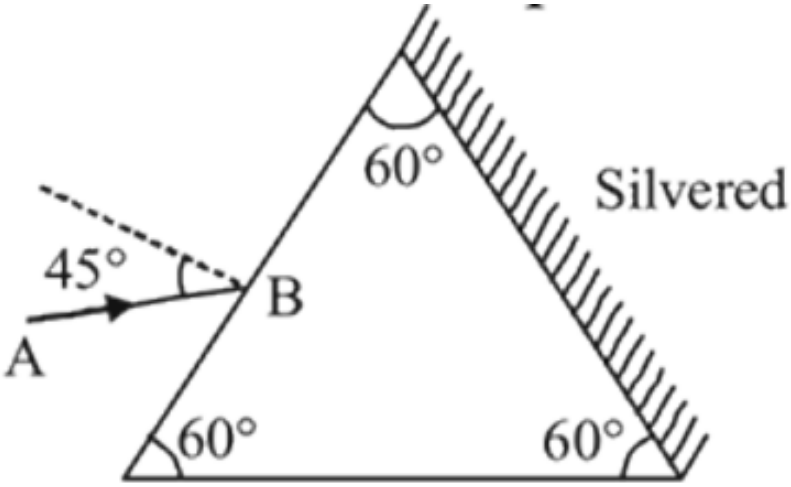
(C) 1.2

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(D) 1.6

Q.8

A equilateral prism is made of a transparent material of refractive index $\sqrt{2}$. A ray oflight AB is incident at 45° as shown. The net deviation in the path of ray when it comes out of prism is



(A) 135°

(B) 120°

(C) 30°

(D) 150°

Q.9

Two beams of red and violet colours are made to pass separately through a prism (angle of the prism is 60°). In the position of minimum deviation, the angle of refraction will be

(A) 30° for both the colours

(B) greater for the violet colour

(C) greater for the red colour

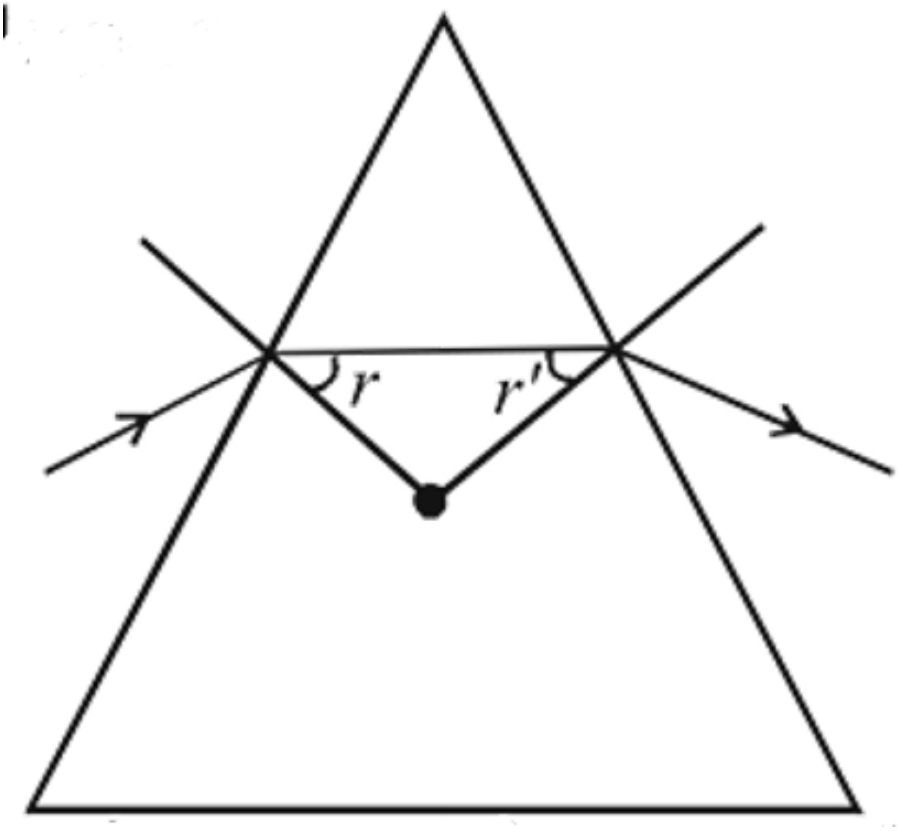
(D) equal but not 30° for both the colours

Q.10

r and r' denote the angles inside an equilateral prism, as usual, in degrees.
Consider that during some time interval from $t = 0$ to $\mathrm{t} = t, r'$
varies with time as $r' = 10 + t^2$. During

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this time r will vary as (assume that r and r' are in degree)



(A) $50 - t^2$

(B) $50 + t^2$

(C) $60 - t^2$

(D) $60 + t^2$

Q.11

A ray of light passes through an equilateral prism such that the angle of incidence is equal to the angle of emergence and the latter is equal to $\frac{3}{4}$ th of angle of prism. The angle of deviation is

(A) 25°

(B) 30°

(C) 45°

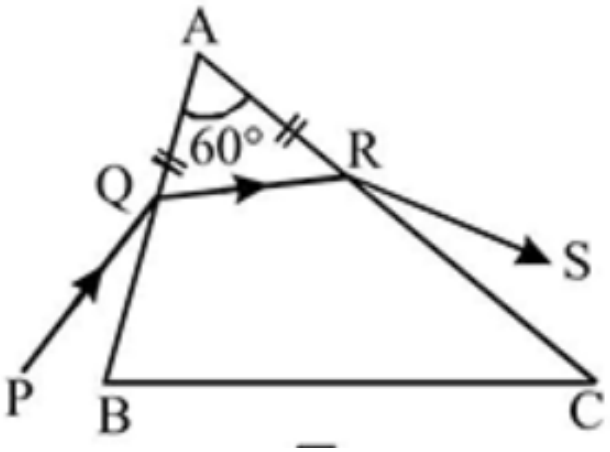
(D) 35°

Q.12

A ray PQ incident on the refracting face BA is refracted in the prism BAC as shown in the figure and emerges from the other refracting face AC as RS such that $AQ = AR$. If the angle of prism $\mathrm{A} = 60^\circ$

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and the refractive index of the material of prism is $\sqrt{3}$, then the angle of deviation of the ray is



(A) 60°

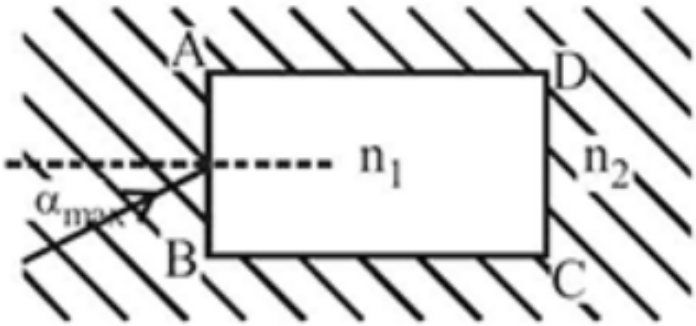
(B) 45°

(C) 30°

(D) None of these

Q.13

A rectangular glass slab $ABCD$ of refractive index n_1 is immersed $x - \cdots_1 \quad n_2$ in water of refractive index n_2 $(n_1 > n_2)$. A ray of light is incident at the surface AB of the slab as shown. The maximum value of the angle of incidence α such that the ray comes out only from the other surface CD is given by



(A) $\sin^{-1}\left[\frac{n_1}{n_2}\cos\left(\sin^{-1}\left(\frac{n_2}{n_1}\right)\right)\right]$

(B) $\sin^{-1}\left[n_1\cos\left(\sin^{-1}\left(\frac{1}{n_2}\right)\right)\right]$

(C) $\sin^{-1}\left(\frac{n_1}{n_2}\right)$

(D) $\sin^{-1}\left(\frac{n_2}{n_1}\right)$

Q.14

A thin prism of angle 15° made of glass of refractive index $\mu_1 = 1.5$ is combined with another prism of

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glass of refractive index $\mu_2 = 1.75$. The combination of the prism produces dispersion without deviation. The angle of the second prism should be

(A) 7°

(B) 10°

(C) 12°

(D) 5°

Q.15

For a prism kept in air it is found that for an angle of incidence 60°, the angle of Prism A, angle of deviation δ and angle of emergence 'e' become equal. Then what will be the refractive index of the prism?

(A) 1.75

(B) 1.78

(C) 1.71

(D) 1.73

Q.16

The diameter of the objective lens of microscope makes an angle β at the focus of the microscope. Further, the medium between the object and the lens is an oil of refractive index n. Then the resolving power of the microscope

(A) increases with decreasing value of n

(B) increases with decreasing value of β

(C) increases with increasing value of $n \sin 2\beta$

(D) increases with increasing value of $\frac{1}{n \sin 2\beta}$

Q.17

A telescope has an objective of focal length 100 cm and an eyepiece of focal length 5 cm. What is the magnifying power of the telescope when the final image is formed at the least distance of distinct vision ?

(A) 20

(B) 24

(C) 28

(D) 32

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Q.18

A microscope is focussed on a mark on a piece of paper and then a slab of glass of thickness $3\ \text{cm}$ and refractive index 1.5 is placed over the mark. How should the microscope be moved to get the mark in focus again ?

(A) $4.5\ \text{cm}$ downward

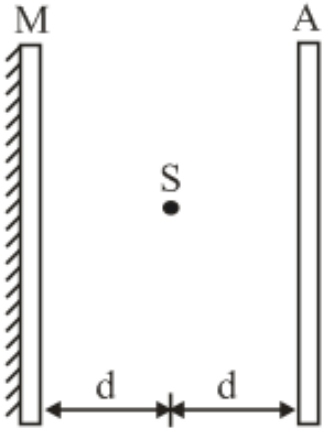
(B) $1\ \text{cm}$ downward

(C) $2\ \text{cm}$ upward

(D) $1\ \text{cm}$ upward

Q.19

A point source of 'S' at a distance d from the screen A produces light intensity I_0 at the centre of the screen. If a completely reflecting mirror M is placed at a distance d behind the source as shown in the figure, the intensity at the centre of the screen changes to :



(A) $\frac{9}{10}I_0$

(B) $\frac{10}{9}I_0$

(C) $\frac{8}{9}I_0$

(D) $\frac{9}{8}I_0$

Q.20

A plano-convex lens has diameter 5 cm and its thickness at the centre is 0.25 cm. If the speed of light inside the lens is 2×10^8 ms^{-1}, then what is the focal length (in cm) of the lens?

(A) 25.35

(B) 25.45

(C) 25.15

(D) 25.25

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Q.1 Answer:

5

Solution:

$M = 30, f_{\mathrm{e}} = 5.0$ cm **and** $D = 25$ cm.

We know that

$M = m_{\mathrm{o}} \times m_e = m_{\mathrm{o}} \left(1 + \frac{D}{f_{\mathrm{e}}}\right)$ or $30 = m_{\mathrm{o}} \left(1 + \frac{25}{5.0}\right)$

which gives $m_{\mathrm{o}} = 5.0$

Q.2 Answer:

-20.7

Solution:

$u = -5.0$ cm, $f_0 = +4.0$ cm **and** $f_{\mathrm{e}} =$ $+6.0$ cm. **For the objective**

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_{\mathrm{o}}}$

Putting $u = -5.0$ cm **and** $f_{\mathrm{o}} = +4.0$ cm **we have**

$\frac{1}{v} = \frac{1}{4} - \frac{1}{5}$

which gives $v = 20$ cm. **Therefore, magnifying power is**

$M = \frac{v}{u}\left(1 + \frac{D}{f_e}\right) = \frac{20}{-5}\left(1 + \frac{25}{6}\right) = -20.7$

The negative sign indicates that final image is virtual.

Q.3 Answer:

green and red lights both emerge from AC

Solution:

Glass is dispersive and the refractive index μ **of glass is related to wavelength** λ **as**

$\mu = A + \frac{B}{\lambda^2}$

where A **and** B **are constants. Since** $\lambda_{\mathrm{R}} > \lambda_{\mathrm{G}} > \lambda_{\mathrm{B}}$, $\mu_{\mathrm{R}} < \mu_{\mathrm{G}} < \mu_{\mathrm{B}}$. **Therefore, blue light is deviated the most and red light is deviated the least. Since blue light is just totally internally reflected at face** AC,

$\sin i_c = \frac{1}{\mu_B}$

where i_c **is the critical angle for blue light. Since** $\mu_{\mathrm{R}} < \mu_{\mathrm{B}}$ **and** $\mu_{\mathrm{G}} < \mu_{\mathrm{B}}$, **the critical angles for red and green lights are greater than that for blue light. Hence red and green light are incident on face** AC **at angles less than their respective critical angle. Therefore, they will not suffer total internal reflection at face** AC **and will emerge out from it. So the correct choice is** (c).

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Q.4 Answer:

3°

Solution:

For a prism with a very small refracting angle A, the deviation is given by (Fig.)

$\therefore$ Deviation produced by the first prism is

$$\delta_1 = (\mu_1 - 1) A_1$$

and that produced by the second prism is

$$\delta_2 = (\mu_2 - 1) A_2$$

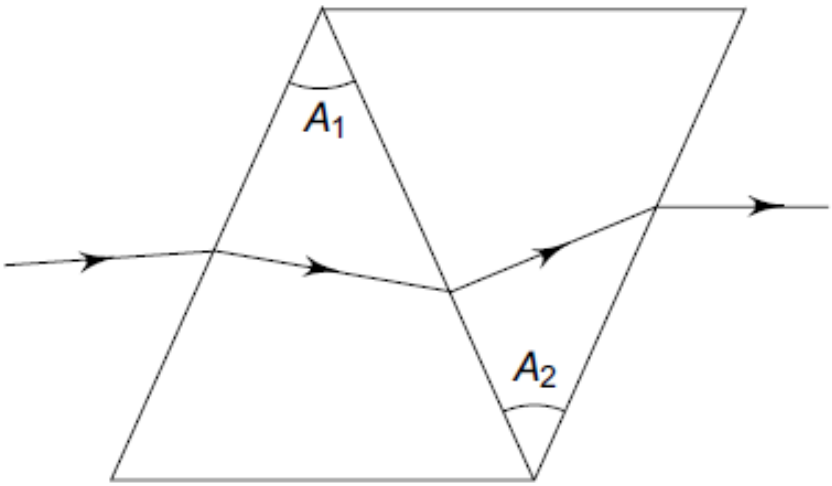
The total deviation will be zero if $\delta_1 + \delta_2 = 0$. The emergent ray will then be parallel to the incident ray (see figure). Thus

$$(\mu_2 - 1) A_2 = -(\mu_1 - 1) A_1$$

The negative sign shows that the refracting angles of the two prisms are in opposite directions. Thus

$$|A_2| = \frac{(\mu_1 - 1) A_1}{(\mu_2 - 1)} = \frac{(1.54 - 1) \times 4^\circ}{(1.72 - 1)} = 3^\circ$$

which is choice (c).



Q.5 Answer:

$\omega d + \omega' d' = 0$

Solution:

Mean angular deviations produced by crown and flint glass prisms respectively are

$$d = (\mu - 1) A \text{ and } d' = (\mu' - 1) A'$$

Their dispersive powers are

$$\omega = \frac{(\mu_v - \mu_r)}{(\mu - 1)} \text{ and } \omega' = \frac{(\mu'_v - \mu'_r)}{(\mu' - 1)}$$

Their angular dispersions respectively are

$$D = (\mu_v - \mu_r) A \text{ and } D' = (\mu'_v - \mu'_r) A'$$

When the prisms are combined, the dispersion by the combination will be zero if

$$D + D' = 0$$

or $(\mu_v - \mu_r) A + (\mu'_v - \mu'_r) A' = 0$

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or $\frac{(\mu_v-\mu_r)}{(\mu-1)}(\mu-1)A+\frac{\left(\mu'_v-\mu'_r\right)}{(\mu'-1)}\left(\mu'-1\right)A'=0$

or $\omega d+\omega' d'=0$**, which is choice (c).**

Q.6 Answer:

$\frac{4\sqrt{2}}{3}$

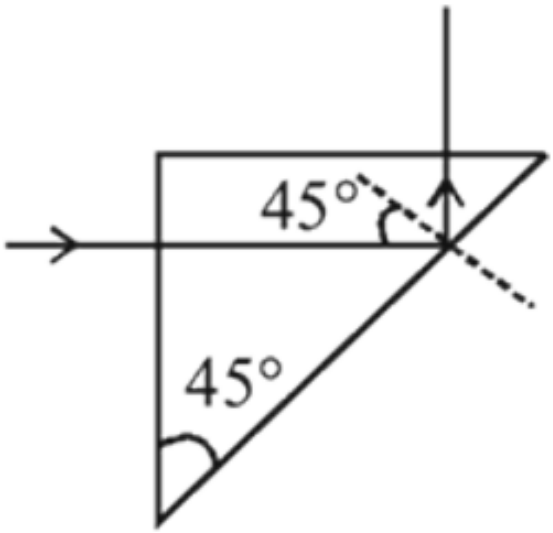
Solution:

ForT.I.R. $45^\circ > C$

$\sin 45^\circ > \sin C$

$\therefore \quad \frac{1}{\sqrt{2}} > \frac{4/3}{n}$

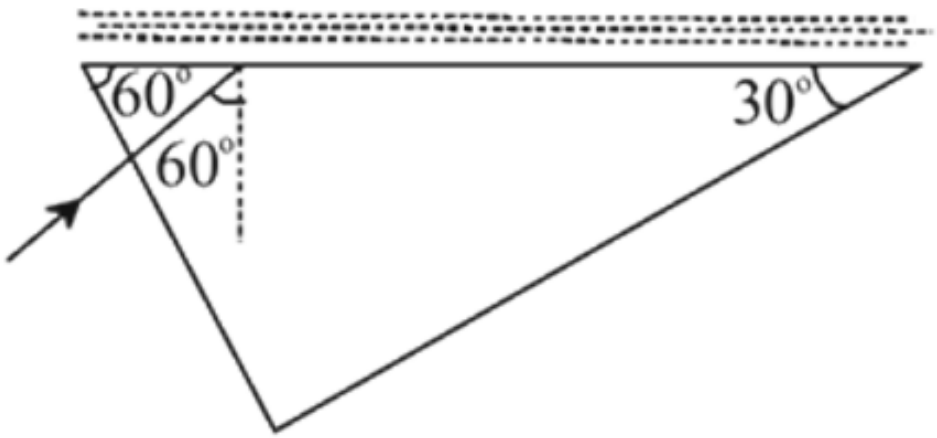
$\therefore \quad n > \frac{4\sqrt{2}}{3}$



Q.7 Answer:

1.3

Solution:



Clearly, $i_c \leq 60^\circ$

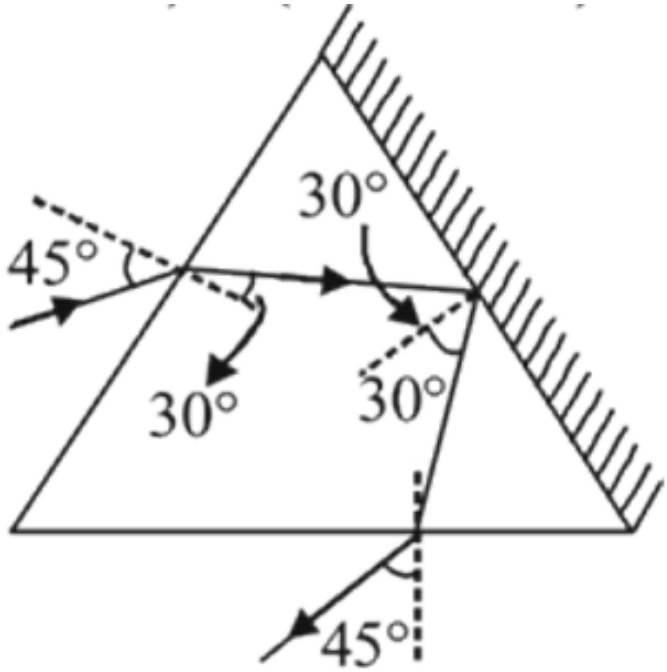
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So, maximum possible value of i_c is 60°. Now, $1\mu_g = \frac{1}{\sin i_c}$
$\frac{\mu_g}{\mu_1} = \frac{1}{\sin i_c}$ **or** $\mu_1 = \mu_g \sin i_c = 1.5 \sin 60^\circ = 1.5 \times \frac{\sqrt{3}}{2}$
$= 1.5 \times 0.866 = 1.299 = 1.3$

Q.8 Answer:

150°

Solution:



$\delta = (45^\circ - 30^\circ) + (180^\circ - 60^\circ) + (45^\circ - 30^\circ)$
$= 150^\circ$ **clockwise.**

Q.9 Answer:

30° for both the colours

Solution:

For minimum deviation the ray in the prism is parallel to the base of the prism. This condition does not depend on the colour (or wave length) of incident radiation. So in both the cases, by geometry, $r = 30^\circ$. So (a) is correct option.

Q.10 Answer:

$50 - t^2$

Solution:

In a prism: $r + r' = A \Rightarrow r = A - r'$
$\therefore r = 60^\circ - \left(10 + t^2\right) = 50 - t^2$

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Q.11 Answer:

30°

Solution:

From the fig. Angle of deviation,

$\delta = i + e - A$

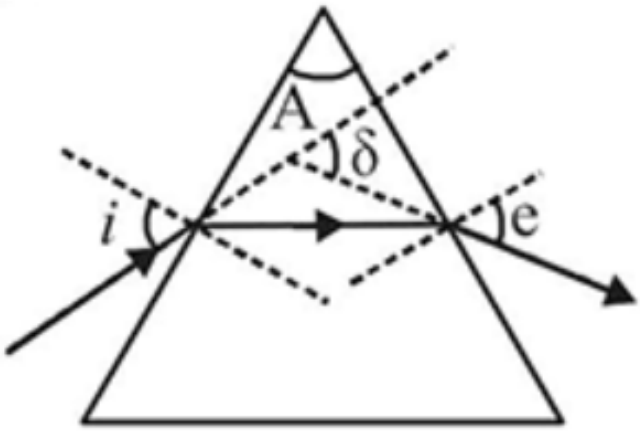
Here, $\mathrm{e} = \mathrm{i}$

and $e = \frac{3}{4}A$

$\therefore \delta = \frac{3}{4}A + \frac{3}{4}A - A = \frac{A}{2}$

For equilateral prism, $\mathrm{A} = 60^\circ$

$\therefore \delta = \frac{60^\circ}{2} = 30^\circ$



Q.12 Answer:

60°

Solution:

Given $\mathrm{AQ} = \mathrm{AR}$ **and** $\angle \mathrm{A} = 60^\circ$

$\therefore \quad \angle \mathrm{AQR} = \angle \mathrm{ARQ} = 60^\circ$

$\therefore \quad \mathrm{r}_1 = \mathrm{r}_2 = 30^\circ$

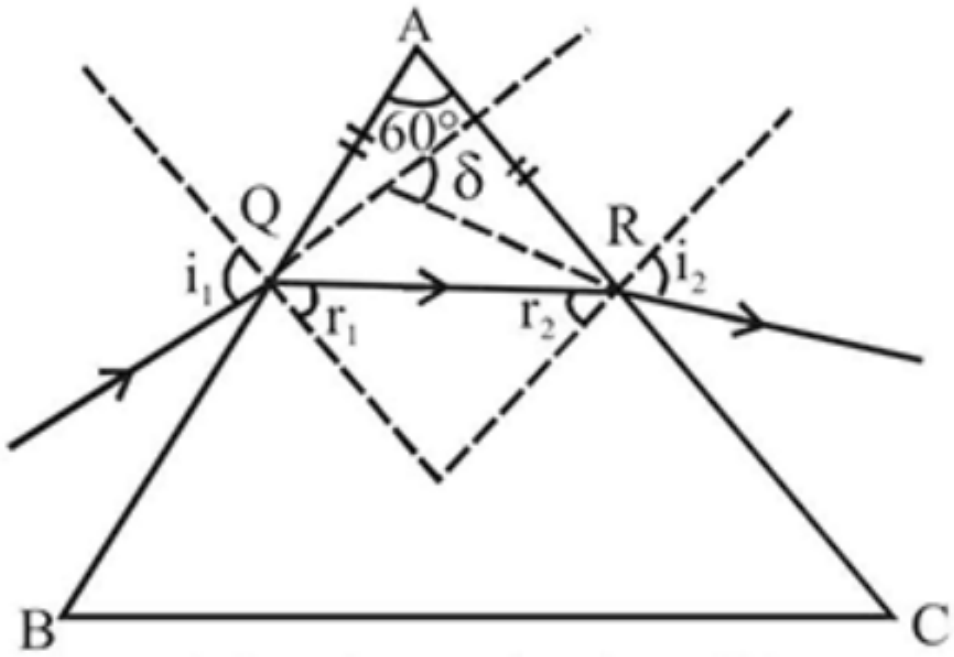
Applying Snell's law on face $\mathrm{AB}.\ \sin \mathrm{i}_1 = \mu \sin \mathrm{r}_1$

$\Rightarrow \sin \mathrm{i}_1 = \sqrt{3} \sin 30^\circ = \sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{2}$

$\therefore \quad \mathrm{i}_1 = 60^\circ$

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Similarly, $i_2 = 60^\circ$ **In a prism, deviation** $\delta = i_1 + i_2 - A = 60^\circ + 60^\circ - 60^\circ = 60^\circ$



Q.13 Answer:

$\sin^{-1}\left[\frac{n_1}{n_2}\cos\left(\sin^{-1}\left(\frac{n_2}{n_1}\right)\right)\right]$

Solution:

The ray will come out from CD **if it suffers total internal reflection at surface** AD**, i.e., it strikes the surface** AD **at critical angle** C **(the limiting case).**

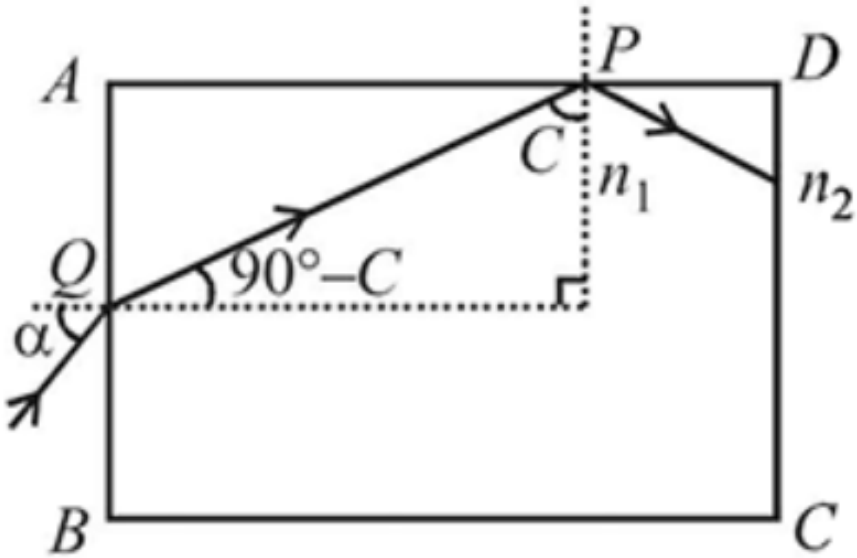
Applying Snell's law at P

$n_1 \sin C = n_2$ **or** $\sin C = \frac{n_2}{n_1}$

Applying Snell's law at Q $n_2 \sin\alpha = n_1 \cos C$

$\Rightarrow \sin\alpha = \frac{n_1}{n_2}\cos\left\{\sin^{-1}\left(\frac{n_2}{n_1}\right)\right\}$

or, $\alpha = \sin^{-1}\left[\frac{n_1}{n_2}\cos\left\{\sin^{-1}\left(\frac{n_2}{n_1}\right)\right\}\right]$



Q.14 Answer:

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10°

Solution:

Deviation = zero

So, $\delta = \delta_1 + \delta_2 = 0$

$\Rightarrow (\mu_1 - 1)\,A_1 + (\mu_2 - 1)\,A_2 = 0$

$\Rightarrow A_2(1.75 - 1) = -(1.5 - 1)15^\circ$

$\Rightarrow A_2 = -\frac{0.5}{0.75} \times 15^\circ$

or $A_2 = -10^\circ$

Negative sign shows that the second prism is inverted with respect to the first.

Q.15 Answer:

1.73

Solution:

Given $\quad i = 60^\circ\ A = \delta = e$

$\delta = i + e - A \Rightarrow \delta = i \quad (\because e = A)$

$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$

Here angle of deviation is $\min .(\because i = e)$

$\mu = \frac{\sin\left(\frac{60^\circ+60^\circ}{2}\right)}{\sin\frac{60^\circ}{2}} = 1.73$

Q.16 Answer:

increases with increasing value of $n \sin 2\beta$

Solution:

Resolving Power of the Microscope is given by $\frac{1}{d} = \frac{2n\sin\theta}{\lambda}$

where n **is refractive index,** θ **is the angle of the cone and** d **is the distance between the object.**

Here, The diameter of the objective lens of microscope makes an angle β **at the focus of the microscope.**

Hence, $\beta = \theta/2$

Hence, resolving power is $\frac{2n \sin 2\beta}{\lambda}$

Q.17 Answer:

24

Solution:

$f_0 = 100\ cm, f_e = 5\ cm$

When final image is formed at least distance of distinct vision (d), then $M = \frac{f_0}{f_e}\left(1 + \frac{f_e}{d}\right) =$

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$\frac{100}{5}\left(1+\frac{5}{25}\right) \quad [\because \mathrm{D}=25 \mathrm{~cm}]$
$\mathrm{M}=20 \times \frac{6}{5}=24$

Q.18 Answer:

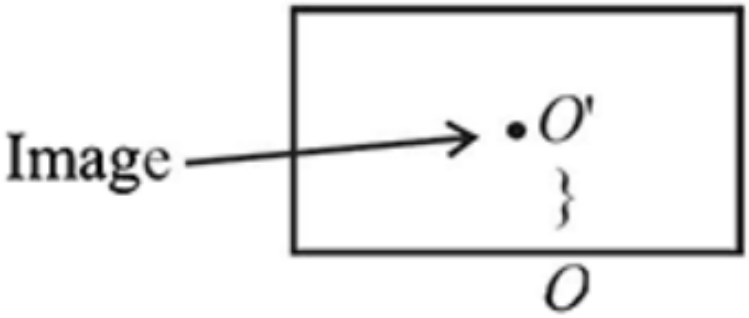
$1 \mathrm{~cm}$ upward

Solution:

In the later case microscope will be focussed for O'. So, it is required to be lifted by distance OO'. OO' = real depth of $\mathrm{O}-$ apparent depth of O.

$=3-\frac{3}{1.5} \quad \left[\mu=\frac{\text{real depth}}{\text{apparent depth}}\right]$

$=3\left[\frac{1.5-1}{1.5}\right]=\frac{3 \times .5}{1.5}=1 \mathrm{~cm}$



Q.19 Answer:

$\frac{10}{9} I_0$

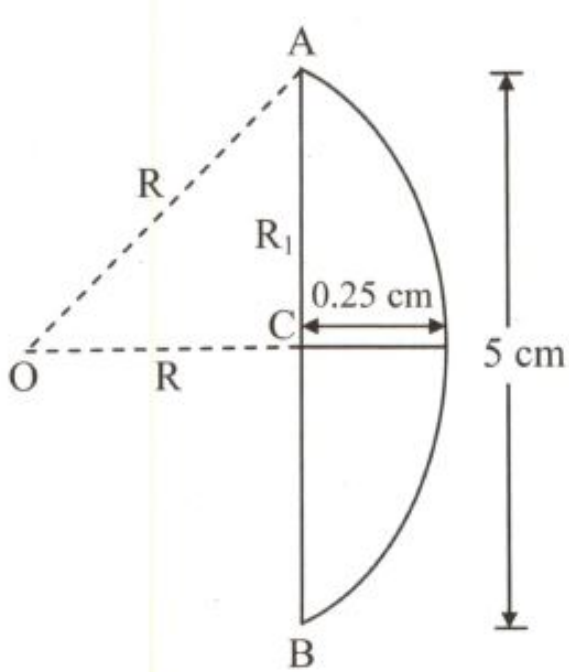
Solution:

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Q.20 Answer:

25.25

Solution:



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R. I. of lens,

$\mu = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = \frac{3}{2}$

From ΔACO, radius of curvature of lens is,

$R^2 = R_1^2 + \left(R - 0.25\right)^2$

$\therefore \quad R^2 = \left(2.5\right)^2 + \left(R - 0.25\right)^2$

$\therefore \quad R^2 = 6.25 + R^2 + 0.0625 - 0.5R$

$\therefore \quad 0.5R = 6.3125$

$\therefore \quad R = 12.625\text{cm}$

$F = \frac{R}{\mu - 1} = \frac{12.625}{\frac{3}{2} - 1} = 25.25 \quad \text{cm}$

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